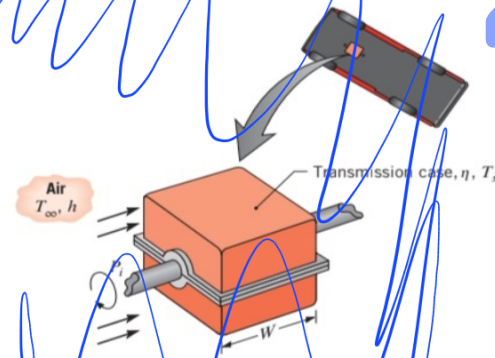


1.14 A wall has inner and outer surface temperatures of 16 and 6°C, respectively. The interior and exterior air temperatures are 20 and 5°C, respectively. The inner and outer convection heat transfer coefficients are 5 and 20 W/m²·K, respectively. Calculate the heat flux from the interior air to the wall, from the wall to the exterior air, and from the wall to the interior air. Is the wall under steady-state conditions?

1.15 The free convection heat transfer coefficient on a thin hot vertical plate suspended in still air can be determined from observations of the change in plate temperature with time as it cools. Assuming the plate is isothermal and radiation exchange with its surroundings is negligible, evaluate the convection coefficient at the instant of time when the plate temperature is 245°C and the change in plate temperature with time (dT/dt) is -0.028 K/s. The ambient air temperature is 25°C and the plate measures 0.4 × 0.4 m with a mass of 4.25 kg and a specific heat of 2770 J/kg·K.

1.16 A transmission case measures $W = 0.30$ m on a side and receives a power input of $P_i = 150$ hp from the engine.

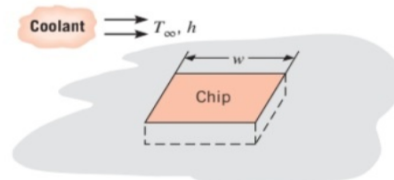


If the transmission efficiency is $\eta = 0.93$ and airflow over the case corresponds to $T_{\infty} = 30^\circ\text{C}$ and $h = 200$ W/m²·K, what is the surface temperature of the transmission? What is the thermal resistance associated with convection?

and surface temperature? What are the consequences of such an event?

1.18 A common procedure for measuring the velocity of an airstream involves the insertion of an electrically heated wire (called a *hot-wire anemometer*) into the airflow, with the axis of the wire oriented perpendicular to the flow direction. The electrical energy dissipated in the wire is assumed to be transferred to the air by forced convection. Hence, for a prescribed electrical power, the temperature of the wire depends on the convection coefficient, which, in turn, depends on the velocity of the air. Consider a wire of length $L = 20$ mm and diameter $D = 0.5$ mm, for which a calibration of the form $V = 6.25 \times 10^{-5} h^2$ has been determined. The velocity V and the convection coefficient h have units of m/s and W/m²·K, respectively. In an application involving air at a temperature of $T_{\infty} = 25^\circ\text{C}$, the surface temperature of the anemometer is maintained at $T_s = 75^\circ\text{C}$ with a voltage drop of 5 V and an electric current of 0.1 A. What is the velocity of the air?

1.19 A square isothermal chip is of width $w = 5$ mm on a side and is mounted in a substrate such that its side and back surfaces are well insulated; the front surface is exposed to the flow of a coolant at $T_{\infty} = 15^\circ\text{C}$. From reliability considerations, the chip temperature must not exceed $T = 85^\circ\text{C}$.



If the coolant is air and the corresponding convection coefficient is $h = 200$ W/m²·K, what is the maximum allowable chip power? If the coolant is a dielectric liquid for which $h = 3000$ W/m²·K, what is the maximum allowable power?

1.20 For a boiling process such as shown in Figure 1.5c

1.6) $\frac{q''}{A} = -k \frac{dT}{dx}$ $50 \text{ mm} \rightarrow .05 \text{ m}$

$$40 \frac{\text{W}}{\text{m}^2} = -k \left(\frac{40 - 20}{.05 \text{ m}} \right)$$

$$40 \frac{\text{W}}{\text{m}^2} = -k \left(400 \frac{\text{K}}{\text{m}} \right)$$

$k = -.1 \frac{\text{W}}{\text{m} \cdot \text{K}}$

1.14) $q'' = h (T_s - T_{\infty})$

Wall: In	c	Out: 6
Air: In: 20°C	Out: 5°C	
h : In: 5 $\frac{\text{W}}{\text{m}^2 \cdot \text{K}}$	Out: 20 $\frac{\text{W}}{\text{m}^2 \cdot \text{K}}$	

$q''(\text{in air} \rightarrow \text{wall}) = 5 \frac{\text{W}}{\text{m}^2 \cdot \text{K}} (16 - 20) = -20 \frac{\text{W}}{\text{m}^2}$

$q''(\text{wall} \rightarrow \text{out air}) = 20 \frac{\text{W}}{\text{m}^2 \cdot \text{K}} (6 - 5) = 20 \frac{\text{W}}{\text{m}^2}$

$q''(\text{wall} \rightarrow \text{in air}) = 5 \frac{\text{W}}{\text{m}^2 \cdot \text{K}} (16 - 20) = -20 \frac{\text{W}}{\text{m}^2}$

yes under steady state

1.15) Convection heat transfer: isothermal, radiation & conduction negligible

Find convection coefficient, plate temp is 245°C

25°C

$\frac{dT}{dt} = -0.028 \frac{\text{K}}{\text{s}}$

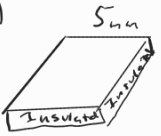
$\frac{dT}{dt} = -.028$
 $q'' = h(T_s - T_\infty)$
 $q'' = -k \frac{dT}{dx} \rightarrow \frac{\text{heat transfer}}{\text{area}} \rightarrow \frac{2770 \frac{J}{kg \cdot K}}{.16 m^2} \cdot (-.028 \frac{K}{s})$

air is $25^\circ C$
 plate is $.4 \times .4$, $m = 4.25 kg$
 specific heat $2770 \frac{J}{kg \cdot K}$

$q'' = h(T_s - T_\infty)$
 $q'' = -k \frac{dT}{dx} \rightarrow \frac{\text{heat transfer}}{\text{area}} \rightarrow \frac{2770 \frac{J}{kg \cdot K} \cdot 4.25 kg}{.16 m^2} \cdot (-.028 \frac{K}{s})$

$q'' = h(T_s - T_\infty) \rightarrow h = \left(\frac{q''}{T_s - T_\infty} \right) \rightarrow \frac{2060.19 \frac{W}{m^2}}{(518 - 298)} = \boxed{9.36 \frac{W}{m^2 \cdot K}}$

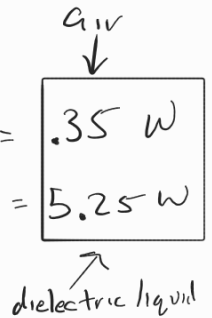
1.19)



$T_\infty = 15^\circ C$, $T = 85^\circ C$
 $h = 200 \frac{W}{m^2 \cdot K}$, what is max power?
 $h = 3000 \frac{W}{m^2 \cdot K}$, power?

$q'' = \frac{W}{m^2} \cdot (\text{Area}) = W \checkmark$

$q'' = h(T_s - T_\infty) \rightarrow q'' = 200 \frac{W}{m^2 \cdot K} (85 - 15) = 14000 \frac{W}{m^2} \cdot (.005 m)^2 = .35 W$
 $q'' = 3000 \frac{W}{m^2 \cdot K} (85 - 15) = 210000 \frac{W}{m^2} \cdot (.005 m)^2 = 5.25 W$



$5 mm \times \frac{10^{-3} m}{1 mm} = .005 m$